

No. of Functional Features Included in the Solution

Date:	15 March 2026
Team ID:	(x x x)
Project Name:	Human Development Index (HDI) Prediction Using Machine Learning
Maximum Marks:	5 Marks

Functional Features Overview:-

This project predicts the **Human Development Index (HDI)** category of a country using Machine Learning. The solution is developed using **Python, Flask, Scikit-learn, Pandas, NumPy, HTML, CSS, and JavaScript**. Users provide development indicators, and the application predicts whether the country falls into **Very High, High, Medium, or Low HDI**.

Functional Features:-

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	User Input Form	Accepts Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and GNI Per Capita from the user.	Frontend (HTML/Flask)	Done	High
2	Data Validation	Checks whether user inputs are valid before prediction.	Backend	Done	High
3	Data Preprocessing	Cleans and prepares input data for the ML model.	Pandas / NumPy	Done	High
4	Machine Learning Prediction	Predicts the HDI category using the trained model.	Scikit-learn	Done	High

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
5	Result Display	Displays the predicted HDI category on the web page.	Flask / HTML	Done	Medium
6	Model Storage	Saves and loads the trained model using model.pkl.	Joblib	Done	Medium
7	Data Visualization	Displays charts and graphs for HDI analysis.	Matplotlib / Seaborn	Done	Medium
8	GitHub Integration	Stores and manages project source code using GitHub.	Git / GitHub	Done	Medium

Feature Summary:-

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6

Additional / Nice-to-Have Features:-

- Responsive user interface using HTML and CSS.
- Easy-to-use prediction form.
- Well-organized project structure.
- Reusable machine learning model.
- Modular Python code.
- GitHub version control.

Features Tested & Verified:-

- User input validation.
- Machine learning prediction accuracy.
- Flask web application functionality.
- Dataset loading.
- Model loading (model.pkl).
- Prediction result display.
- Error handling for invalid input.
- GitHub repository verification.

Feature Category Breakdown :-

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Input form, Prediction result page
2	Backend / Logic	3	Data preprocessing, Prediction engine, Validation
3	Database / Storage	1	CSV dataset, Trained model (model.pkl)
4	API / Integration	1	Flask application integration
5	Security / Authentication	1	Input validation and error handling
